Package ‘GR2MSemiDistr’

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Type Package

Title Semi-Distributed GR2M Model for Large-Sample Hydrological Applications

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Description Implements a semi-distributed adaptation of the GR2M monthly hydrological model for large-sample and operational hydrological modeling. The model uses regional parameters and subbasin-level forcing data to simulate water balance components such as precipitation, evapotranspiration, runoff, and discharge. It supports warm-up periods, initial states, output saving, regional parameter correction, and performance evaluation through various objective functions. The routing scheme is based on subbasin structure and spatial relationships derived from a DEM using WhiteboxTools. Designed for flexible integration in research, water resource assessments, and operational hydrological systems in complex basins.

License GPL (>= 2)

Encoding UTF-8

Depends R (>= 4.0)

Imports airGR, hydroGOF, Matrix, terra, lubridate, tictoc, sf, rtop, whitebox, exactextractr

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NeedsCompilation no

Contents

`Create_Forcing_Inputs` *Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.*

Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = NULL
)
```

Arguments

`Subbasins`	A SpatVector object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).
`Precip`	A SpatRaster object with monthly precipitation data in [mm/month].
`PotEvap`	A SpatRaster object with monthly potential evapotranspiration data in [mm/month].
`Qobs`	Optional. A vector with observed streamflow in [m³/s] at the basin outlet.
`DateIni`	Start date of the output time series in 'mm/yyyy' format.
`DateEnd`	End date of the output time series in 'mm/yyyy' format.
`IniGrids`	Initial date of the gridded precipitation and evapotranspiration datasets in mm/yyyy format.

`Save`	Logical. If TRUE, saves the resulting data frame to a .txt file in the './Inputs' directory. Default is FALSE.
`Update`	Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.
`Members`	Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.

Value

A data frame containing the model inputs in the airGR format, with columns:

DatesR Monthly dates

P_x Mean precipitation for subbasin x, where x is the COMID

E_x Mean potential evapotranspiration for subbasin x, where x is the COMID

Q Observed streamflow [m³/s], if provided

References

Aybar C, Fernández C, Huerta A, Lavado W, Vega F, Felipe-Obando O. (2020). Construction of a high-resolution gridded rainfall dataset for Peru from 1981 to the present day. Hydrological Sciences Journal, 65(5), 770–785. doi:10.1080/02626667.2019.1649411

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). Water, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Load example data
data(cat)         # Subbasin geometries (SpatVector)
data(pisco_pr)    # Monthly precipitation raster [mm/month]
data(pisco_pe)    # Monthly potential evapotranspiration raster [mm/month]
data(qobs)        # Observed streamflow [m³/s]

# Define simulation period and grid start date
start_date <- "01/1981"
end_date   <- "12/2016"
ini_grids  <- "01/1981"

# Create input data for the model
model_inputs <- Create_Forcing_Inputs(
  Subbasins = cat,
  Precip = pisco_pr,
  PotEvap = pisco_pe,
  Qobs = qobs,
  DateIni = start_date,
  DateEnd = end_date,
  IniGrids = ini_grids,
  Save = FALSE
)

# Create figures to visualize the inputs
```

```
dates <- as.Date(model_inputs$DatesR)

# Plot average precipitation across subbasins
P_vars <- grep("^P_", names(model_inputs), value = TRUE)
avg_P <- rowMeans(model_inputs[, P_vars], na.rm = TRUE)
plot(dates, avg_P, type = "l", col = "blue", lwd = 1.5,
     ylab = "Precipitation [mm/month]", xlab = "Date",
     main = "Mean Precipitation across Subbasins")

# Plot average potential evapotranspiration across subbasins
E_vars <- grep("^E_", names(model_inputs), value = TRUE)
avg_E <- rowMeans(model_inputs[, E_vars], na.rm = TRUE)
plot(dates, avg_E, type = "l", col = "orange", lwd = 1.5,
     ylab = "PET [mm/month]", xlab = "Date",
     main = "Mean Potential Evapotranspiration across Subbasins")

# Plot observed streamflow at outlet (if available)
if ("Q" %in% names(model_inputs)) {
  plot(dates, model_inputs$Q, type = "l", col = "darkgreen", lwd = 1.5,
       ylab = "Observed Streamflow [m³/s]", xlab = "Date",
       main = "Observed Streamflow at Outlet")
}
```

`Optim_GR2MSemiDistr` *Parameter optimization for a semi-distributed GR2M model using the SCE-UA algorithm.*

Description

Optimizes the GR2M monthly water balance model parameters (X1, X2, fp, fe) for multiple calibration regions using the SCE-UA algorithm. The model is run in a semi-distributed setup over multiple subbasins grouped by region. Users can select from various objective functions (e.g., NSE, KGE, RMSE, and linear combinations), including composite metrics. Parameter bounds must be provided, and regions can be excluded from calibration.

Usage

```
Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = 1500,
  Optimization = "OF1",
  WarmUp = 36,
  No.Optim = NULL,
  w1 = 0.6,
  w2 = 0.3,
  w3 = 0.2
)
```

Arguments

`Data`	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and Q (used for calibration).
`Subbasins`	A SpatVector object containing the geometries of subbasins. It must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).
`RunIni`	Simulation start date in the format "mm/yyyy".
`RunEnd`	Simulation end date in the format "mm/yyyy".
`Parameters`	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
`Parameters.Min`	Numeric vector of length 4, specifying lower bounds for optimization in the following order: c(X1, X2, fp, fe).
`Parameters.Max`	Numeric vector of length 4, specifying upper bounds for optimization in the same order as ‘Parameters.Min‘.
`Optimization`	Character. Objective function to optimize. Available options include: **"OF1"** Kling-Gupta Efficiency (KGE). **"OF2"** Nash-Sutcliffe Efficiency (NSE). **"OF3"** NSE applied to log-transformed flows (log-NSE, Pushpalatha 2012). **"OF4"** Root Mean Square Error (RMSE). **"OF5"** Percent Bias (PBIAS). **"OF6"** FDC Bias (FDC Bias), based on flow duration curves. **"OF7"** Pearson correlation coefficient (r). **"OF8"** Composite objective: w1 × KGE + w2 × NSE - w3 × RMSE. **"OF9"** Composite objective: w1 × KGE + w2 × log-NSE - w3 × RMSE. **"OF10"** Composite objective: w1 × KGE.km + w2 × KGE.lf - w3 × RMSE.
`No.Optim`	Character vector (optional). Region names to exclude from the optimization (parameters will be kept fixed).
`w1`	Numeric. Weight for the first component in composite objective functions (OF8, OF9, OF10). Default is 0.6.
`w2`	Numeric. Weight for the second component in composite objective functions. Default is 0.3.
`w3`	Numeric. Weight for the third component in composite objective functions. Default is 0.2.
`Max.Functions`	Integer. Maximum number of function evaluations in the optimization. Default is 1500.

Value

A list with the following elements:

Parameters Data frame of optimized parameters per region (columns: Region, X1, X2, fp, fe).

OF Final value of the selected objective function.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Load preprocessed input data
data(cat)     # Subbasins (SpatVector with COMID and Region)

# Define initial parameters for each region
param_init <- data.frame(
  Region = unique(cat$Region),
  X1 = rep(500, length(unique(cat$Region))),
  X2 = rep(1.5, length(unique(cat$Region))),
  fp = rep(1.0, length(unique(cat$Region))),
  fe = rep(1.0, length(unique(cat$Region)))
)

# Optimize parameters using the KGE criterion (OF1)
result <- Optim_GR2MSemiDistr(
  Data = inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  Parameters = param_init,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = c(2000, 10, 1.2, 1.2),
  WarmUp = 12,
  Max.Functions = 1000,
  Optimization = "OF1"  # KGE
)

# Extract results
best_params <- result$Parameters
final_score <- result$OF

# Plot observed vs simulated discharge at the outlet
model <- Run_GR2MSemiDistr(
  Data = inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  Parameters = best_params,
  WarmUp = 12
)

if (!is.null(model$SINK)) {
  dates <- as.Date(model$Dates)
  plot(dates, model$SINK$sim, type = "l", col = "blue", lwd = 2,
       xlab = "Date", ylab = "Discharge [m³/s]",
       main = "Simulated vs Observed Discharge (Outlet)")
  lines(dates, model$SINK$obs, col = "darkgreen", lwd = 2, lty = 2)
  legend("topright", legend = c("Simulated", "Observed"),
         col = c("blue", "darkgreen"), lty = c(1, 2), lwd = 2, bty = "n")
}
```

`Routing_GR2MSemiDistr` *Routing of simulated discharges for each subbasin using a weighted flow accumulation approach.*

Description

This function routes monthly simulated streamflows from a semi-distributed GR2M model using a flow accumulation approach based on D8 flow directions. The routing network is derived from a digital elevation model (DEM) via WhiteboxTools, or optionally from a precomputed transfer matrix. Each subbasin contributes its outflow as a weight that is accumulated along the flow direction grid to downstream subbasins.

Usage

```
Routing_GR2MSemiDistr(
  Model,
  Subbasins,
  Dem,
  RouIni = NULL,
  RouEnd = NULL,
  TransferMatrixFile = NULL,
  res = 250,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

`Model` List containing model results, typically from 'Run_GR2MSemiDistr'. Must include:

- **QS** Matrix of simulated streamflows [m³/s], one column per subbasin.
- **Dates** Vector of dates corresponding to the rows of QS.

`Subbasins` A 'SpatVector' object with the subbasin boundaries. Must include attributes:

- **COMID** Integer ID of each subbasin.
- **Region** Region code or name.

`Dem` A 'SpatRaster' representing the digital elevation model. Required unless 'TransferMatrixFile' is used.

`RouIni` Character. Start date for routing in "mm/YYYY" format. If NULL, uses the start of the simulation.

`RouEnd` Character. End date for routing in "mm/YYYY" format. If NULL, uses the end of the simulation.

`TransferMatrixFile` Optional file path (.rds) to a precomputed transfer matrix. If provided, 'Dem' and WhiteboxTools are not used.

`res` Numeric. Desired spatial resolution in meters for resampling the DEM (default is 250). Converted to degrees if needed.

`Save` Logical. If TRUE, writes the routed discharges (QR) to a tabular .txt file in the './Outputs' directory. Default is FALSE.

`Update` Logical. If TRUE, deletes previous month's output and replaces it with the latest. Useful for incremental runs. Default is FALSE.

Value

A list with the following elements:

QR Matrix of routed streamflows [m³/s] per subbasin and time step.

Dates Vector of routing time steps.

COMID Vector of subbasin COMIDs (column names of QR).

TransferMatrix Sparse matrix used for flow routing (rows = outlets, columns = sources).

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). Water, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Load required data
data(model)  # Output from Run_GR2MSemiDistr
data(cat)    # Subbasin polygons (SpatVector)
data(dem)    # Digital elevation model (SpatRaster)

# Run routing using DEM and WhiteboxTools
routed <- Routing_GR2MSemiDistr(
  Model = model,
  Subbasins = cat,
  Dem = dem,
  RouIni = "01/1981",
  RouEnd = "12/2016",
  Save = FALSE
 )

# Select COMID of the subbasin to plot
idx <- which(model$COMID == model$COMID[1]) # Change index as needed
dates <- as.Date(model$Dates)

plot(dates, routed$QR[, idx], type = "l", col = "darkblue", lwd = 2,
     xlab = "Date", ylab = "Discharge [m³/s]",
     main = paste("Routed Discharge - Subbasin", routed$COMID[idx]))
```

Run_GR2MSemiDistr *Run the GR2M model for multiple subbasins*

Description

This function executes the GR2M monthly water balance model across multiple subbasins using a semi-distributed approach. It applies region-specific parameters and correction factors for precipitation and evapotranspiration, and optionally saves the results as text files.

Usage

```
Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

`Data`	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and optionally Q.
`Subbasins`	A SpatVector object containing the geometries of subbasins. It must include attributes 'COMID' (unique subbasin ID) and 'Region' (region name/code).
`RunIni`	Simulation start date in the format 'mm/yyyy'.
`RunEnd`	Simulation end date in the format 'mm/yyyy'.
`Parameters`	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
`WarmUp`	Optional number of months to discard from the beginning of the simulation as warm-up. Default is NULL.
`IniState`	Optional list of initial states for each subbasin. Default is NULL.
`Save`	Logical. If TRUE, output time series will be saved as text files in the "Outputs/" folder.
`Update`	Logical. If TRUE and Save = TRUE, it will remove previous month's output files before saving the current ones.

Value

A list with the following elements:

PR Matrix of total precipitation [mm/month] for each subbasin

AE Matrix of actual evapotranspiration [mm/month] for each subbasin

SM Matrix of production store level [mm/month] for each subbasin

SM Matrix of percolation [mm/month] for each subbasin

RU Matrix of runoff [mm/month] for each subbasin

QS Matrix of flow [m3/s] (not routed) for each subbasin

Dates Date vector for the simulation period

COMID Vector of subbasin COMIDs

EndState List of final state variables for each subbasin

SINK Optional. Data frame of simulated and observed outlet discharge, if observed Q is provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). Water, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Load example data
data(cat)

# Define region parameters
param_df <- data.frame(
  Region = unique(cat$Region),
  X1 = 350,
  X2 = 1.2,
  fp = 1.0,
  fe = 1.0
)

# Run the semi-distributed GR2M model
model <- Run_GR2MSemiDistr(
  Data = inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  Parameters = param_df,
  Save = FALSE
)

# Select COMID of the subbasin to plot
target_comid <- model$COMID[1]  # Change index as needed
idx <- which(model$COMID == target_comid)
dates <- as.Date(model$Dates)

# Plot model outputs for selected subbasin
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

plot(dates, model$PR[, idx], type = "l", col = "blue",
     main = paste("Precipitation (PR) - COMID", target_comid),
     xlab = "Date", ylab = "mm")

plot(dates, model$AE[, idx], type = "l", col = "orange",
     main = paste("Actual Evapotranspiration (AE) - COMID", target_comid),
     xlab = "Date", ylab = "mm")

plot(dates, model$SM[, idx], type = "l", col = "green4",
     main = paste("Production Store (SM) - COMID", target_comid),
     xlab = "Date", ylab = "mm")

plot(dates, model$PC[, idx], type = "l", col = "purple",
     main = paste("Percolation (PC) - COMID", target_comid),
     xlab = "Date", ylab = "mm")

plot(dates, model$RU[, idx], type = "l", col = "darkred",
```

```
     main = paste("Runoff (RU) - COMID", target_comid),
     xlab = "Date", ylab = "mm")

plot(dates, model$QS[, idx], type = "l", col = "blue",
     main = paste("Simulated Discharge (QS) - COMID", target_comid),
     xlab = "Date", ylab = "m³/s")

# Optional: compare simulated vs observed outlet discharge
if (!is.null(model$SINK)) {
  par(mfrow = c(1, 1))
  plot(dates, model$SINK$sim, type = "l", col = "blue", lwd = 2,
       xlab = "Date", ylab = "Discharge [m³/s]",
       main = "Simulated vs Observed Discharge (Outlet)")
  lines(dates, model$SINK$obs, col = "darkgreen", lwd = 2, lty = 2)
  legend("topright", legend = c("Simulated", "Observed"),
         col = c("blue", "darkgreen"), lty = c(1, 2), lwd = 2, bty = "n")
}
```

Index